

TimeLightHub

A Composable Photonic Computer That Thinks in Light

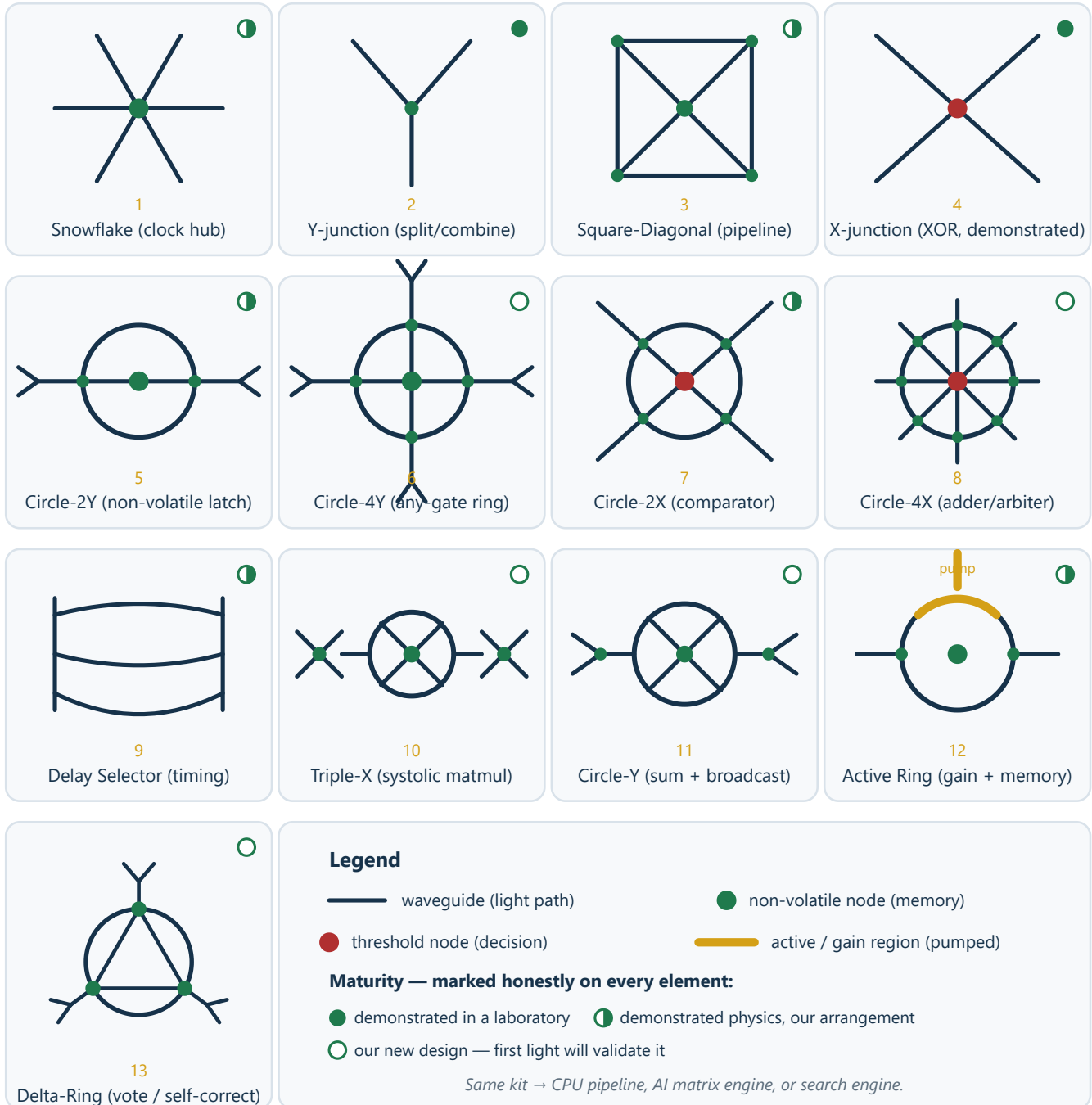
Complete Research — Samer Beyrouti, Inventor & Architect · ORCID 0009-0003-8565-2605

June 2026 · Patent Pending — UK application GB2615417.9 (filed 18 June 2026)

Connect@TimeLightHub.com · +961 3 097 151

TimeLightHub — The 13 Photonic Elements

a composable kit: the same elements, arranged differently, become different machines



A Composable Photonic Computer That Thinks in Light

Samer Beyrouti — Inventor & Architect June 2026 · Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151

We set out to answer a simple question: what would a computer look like if it were built from light instead of electricity — if it computed almost without heat, remembered without power, and could be reshaped, by color, into whatever machine the task required?

This paper is the answer. It introduces TimeLightHub: a computer made of thirteen photonic elements that compose, like the sections of an orchestra, into a CPU, an AI engine, or a search engine — all from the same light.

1. Why light

Every electronic computer pays the same four taxes. It **charges and discharges billions of tiny capacitors** every second, just to flip its switches. It **heats its own wires** with the current it pushes through them — which is why a single AI chip can draw seven hundred watts and demand constant cooling. It **refreshes its memory** thousands of times a second, spending power just to not forget. And above all, it **spends most of its energy not computing, but moving data** — dragging an AI model's weights across the chip for every word it generates.

Light pays none of these in the path where it computes. A photon carries information without charging a capacitor. It travels a waveguide without driving current through resistance. Its memory, stored as a stable molecular state, keeps itself with no refresh. And if the data it needs is already stored as light, in place, it never has to be moved at all.

TimeLightHub is built on that observation — and on one more: **the answer to 7×8 has always been 56**. We do not need to *compute* it every time. We can write it, once, into a permanent structure of light and simply **navigate** to it. From that single idea — computation as navigation through pre-written light — grows an entire architecture.

2. The core idea: navigation, not calculation

At the heart of TimeLightHub is the **OmniCube**: a small, permanent volume of light that stores the result of every basic operation for every possible byte, and returns any one of them in a single step. A complete byte-arithmetic unit holds about 655,360 answers — written once, stable for centuries, drawing no power to remember.

The same principle reaches further. The OmniCube that stores arithmetic also drives the multiply-and-add at the heart of artificial intelligence, image processing, and science. **We pre-store what is finite, and accelerate what is not** — a discipline that keeps every claim in this work grounded in what light can actually do.

3. What TimeLightHub gives you

- **Far less energy per answer.** Computing in light avoids the four electronic taxes inside the compute path — and for AI, where most energy is spent moving model weights, TimeLightHub keeps the weights **where the light reads them**, so they never move. The whole advantage reduces to one measurable number — the energy to read one stored weight — and published photonic physics says that number can land where the advantage is *tens of times*, not percent. Our first experiment measures it.
- **It runs cool.** The light path itself produces almost no heat. What heat remains lives at the edges — the lasers and the sparse amplifying rings — where it is budgeted, placed, and carried away by design.
- **It never forgets.** Every memory tier is non-volatile. Data is held as a stable molecular state that survives power loss and lasts for years to centuries. The machine wakes instantly, with its memory intact.
- **It corrects itself.** A three-fold element votes among three copies of a value and discards errors — fault tolerance built into the geometry, not bolted on.
- **It is composable.** Thirteen elements, reconfigured by color, become different machines. One device, many computers.
- **A handful of parts.** Light replaces billions of transistors with a small set of structures — fewer parts, fewer failures.

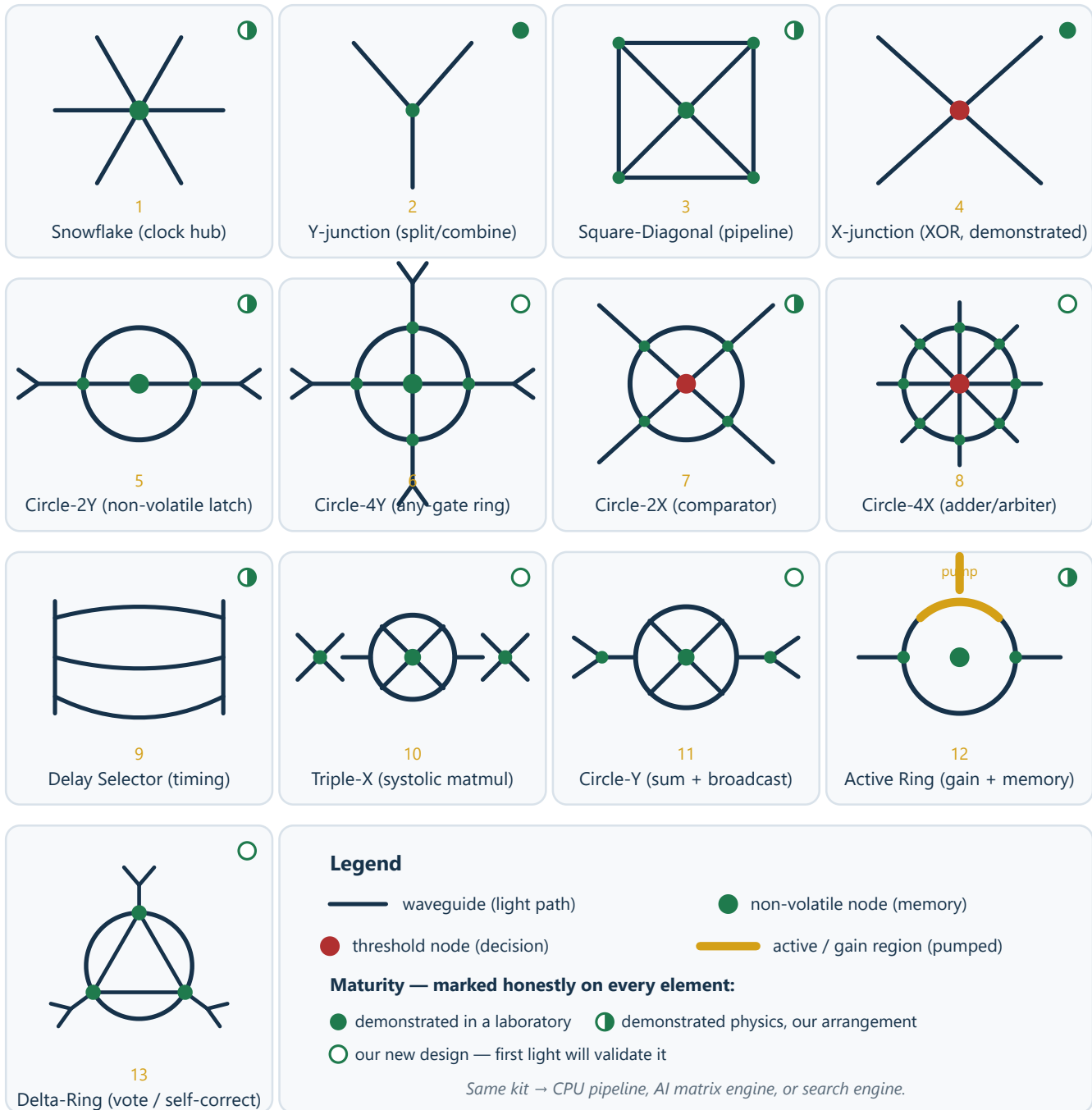
4. The thirteen elements

TimeLightHub is not a fixed chip. It is a **kit** — thirteen photonic elements, each a small structure of waveguides, rings, and light-switchable nodes. They are the instruments; the architecture is the music

they make together.

TimeLightHub — The 13 Photonic Elements

a composable kit: the same elements, arranged differently, become different machines



Each element plays a part:

- **The conductor — the Snowflake (1).** A six-armed hub that sends a single pulse of light, the "heartbeat," to every element at once along paths of equal length — a near-zero-skew clock, by construction, made of light.

- **The memory — the Circle-2Y latch (5) and the Active Ring (12).** A ring of light with a light-switchable node at its center holds a bit *with the power off* — a non-volatile memory made of light. The device physics behind it has been demonstrated in research laboratories; what is ours is making it a **composable element of a computer**, not a laboratory curiosity. The Active Ring adds gain: it is memory and amplifier in one, keeping signals bright and clean as they travel.
- **The logic — the X-junction (4), Square-Diagonal (3), and the Circle gates (6, 7, 8).** Crossings of light that perform the full set of logical operations, the fastest of them already demonstrated in the laboratory at a sixth of a trillionth of a second.
- **The routing — the Y-junction (2) and Delay Selector (9).** Light that splits, combines, broadcasts, and is timed to the picosecond by the length of its own path.
- **The compute fabric — the Triple-X cell (10) and the Circle-Y cell (11).** The engines of artificial intelligence: a grid that multiplies matrices with its **weights held in place**, and a cell that sums, activates, and broadcasts — the exact dataflow of a neural network layer.
- **The self-correction — the Delta-Ring (13).** A triangle of light inside a circle that takes three copies of a value and **votes** — discarding noise and error by agreement, the way the most reliable machines in aerospace do, but built directly into the optics.

We mark each element honestly, and the catalog shows it: some are **already demonstrated** in laboratories, most rest on **demonstrated physics in a new arrangement**, and some are **our own new designs** whose first validation will come at first light. We know exactly which is which — and that knowledge is the build plan.

The complete physics of each element — the materials, the geometries, the fabrication — is shared with research and manufacturing partners under agreement (Section 8).

5. The orchestra: one kit, many machines

The power of TimeLightHub is that **arrangement is function**. Wire the elements in a line and you have a processor's pipeline. Spread them wide, each on its own color of light, and you have a massively parallel engine. Tile them in a grid and you have an artificial-intelligence matrix engine. Branch them into a tree and you have a search engine. Loop them back and you have a machine that iterates and simulates.

A CPU, a GPU, and a search engine are not three different chips here. **They are the same thirteen elements, arranged differently — and the arrangement can be changed by light.** The TimeLightHub programming language, LUMINA, is the score: it compiles a program into the colors and timings that arrange the orchestra for the task at hand. *Same hardware, different music.*

6. Why the energy is so low — the mechanism

The energy advantage is not a hope; it is a consequence of where the energy goes.

In an electronic computer, energy is spent **charging capacitors** (the largest cost), **heating wires**, **refreshing memory** so it does not forget, and above all **moving data**. TimeLightHub removes each one from the compute path: light computes without charging capacitors and without heating wires; its memory is stable and needs no refresh; and in its AI engine the **model's weights live inside the array, where the light multiplies them — so they never move**. Moving data is the single largest energy cost in modern AI, and TimeLightHub is built to make it nearly disappear.

What energy remains sits at the **edges** — the lasers that make the light, the modulators that write the data onto it once at the input, and the pumps that feed the sparse amplifying rings. Paid at the boundary and at intervals, not per operation, everywhere.

The optical multiply itself is so efficient it has been performed, in published laboratory work, with **less than one photon of light** — efficiency at the level of the laws of physics. The result is a machine designed to answer each question for a small fraction of the energy of today's hardware, and to do it cool and quiet. And we are honest about the boundary of the claim: the win is **energy per answer**, not raw speed — a warehouse of top GPUs will still outrun a first TimeLightHub on sheer throughput. It will just pay many times the electricity bill to do it.

7. From idea to first light

We are a research program with a clear, honest path. The architecture is complete on paper; the physics of every element rests on published science; and the whole of it reduces to a single, decisive first experiment: **build one light-switchable memory cell and measure how fast and how cheaply it reads and writes**. That one result — modest in cost, months in time, publishable on its own — turns the architecture from design into evidence, and it is the first note of the orchestra.

From there the path is staged and deliberate: one byte and its first lookup, a single processing node, one seven-cell core that plays a short piece in tune, and then the full Unit. Two engineering questions remain genuinely open — how deep an analog value can cascade before precision fades, and the energy budget of the amplifying rings — and each has a designed answer and a stage on the roadmap that measures it. We would rather show a quartet that plays beautifully than promise a symphony on paper.

8. Working with us

This paper is an invitation. It presents the vision, the architecture, and the advantages in full — and it deliberately keeps the deepest engineering (the element physics, the fabrication recipes, the measured design parameters) for partners who join us under agreement. We are seeking:

- **A photonics laboratory or foundry** to build the first memory cell and the first node.
- **Research and funding partners** who see what a cool, non-volatile, composable optical computer could mean for energy-efficient AI.

The intellectual property is protected; the documentation is comprehensive; and the first experiment is ready to run.

9. The vision

A computer of a few dozen parts instead of billions. That never loses your data. That barely warms. That corrects its own errors. That answers each question for a fraction of the energy — and can be reshaped, by light, into whatever machine you need.

The science of each piece is real. The architecture is complete. The only question left is not *whether* a computer can think in light — but *who builds the first one*.

Samer Beyrouti — TimeLightHub — June 2026 — Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151 Full technical documentation and the element-physics dossier available to partners under agreement.

The OmniCube

A Library of Answers, Written in Light

Samer Beyrouti — Inventor & Architect · TimeLightHub · June 2026

The answer to 7×8 has always been 56. It was true before the first computer, before the first human, before the universe began. So why compute it at all? Why not simply write it down — once, in light — and reach for it when needed?

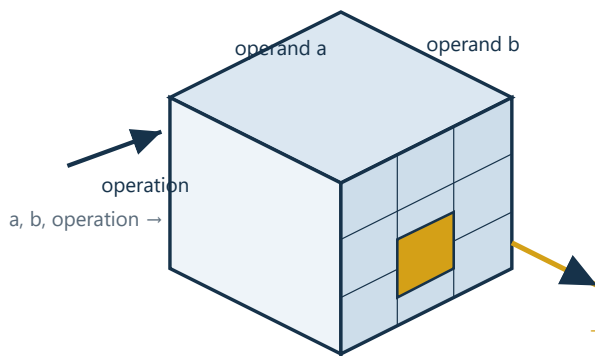
1. The idea: don't compute, navigate

Every other computer *calculates* its answers, switching millions of gates to grind out a result it will grind out again the next time. TimeLightHub does something different. It **stores the answer once** — permanently, in a small structure of light — and then simply **navigates to it**.

That structure is the **OmniCube**. Think of it as a three-dimensional library: give it two numbers and an operation — a , b , and "multiply," say — and those three coordinates point to a single shelf where the answer already waits. The light goes there and brings it back, in one step.

The OmniCube — a library of answers in light

Don't compute — navigate to the stored answer



The domains — swappable libraries of knowledge

Math + · − × ÷ √	AI activation GELU · Softmax	Cipher AES S-box
Pixel image filters	Color RGB ↔ spaces	Text ASCII · UTF-8
Routing network paths	Gradient for AI training	Trig sin · cos · √

...and more — each a module, added like a new cartridge of knowledge.

655,360 answers, written once into light — stable for centuries, no power to keep.

One engine, two gifts

Direct answers for bounded questions (arithmetic, ciphers, activations) — and the **multiply-add engine** that powers AI and science. We pre-store what is finite, and accelerate what is not — never pretending to store infinity.

A complete arithmetic OmniCube holds about **655,360 answers** — every result of every basic operation on every pair of bytes. It is written once, at the factory, into a permanent light-sensitive material that holds its pattern **for centuries and needs no power to remember**. Mathematical truth does not change, so it never needs updating.

And when the numbers grow wider than a byte — 32 bits, 64 bits — the byte-answers **compose**: the Circle-4X element carries from one byte to the next, the way a child carries a digit in long arithmetic. Composition takes steps, and we count them honestly; but each step is still a single navigation, not a thousand switching gates. The library does the knowing; the elements do the carrying.

2. The domains: a growing library of knowledge

Arithmetic is only the first volume. The same principle — *store what is knowable, then navigate to it* — applies to far more than numbers. Each kind of knowledge becomes its own **domain module**, a swappable library of pre-written answers:

- **Math** — add, subtract, multiply, divide, and more.
- **AI activation** — the exact functions (GELU, Softmax) that artificial intelligence runs on, every layer. *Storing these as byte-tables is not our speculation: published AI research already runs them as integer lookup tables with negligible loss of accuracy.*
- **Cipher** — the substitution at the heart of AES encryption.

- **Pixel** — every image filter, for every pixel value.
- **Color** — conversions between every color space.
- **Text** — letters, cases, scripts, encodings.
- **Routing** — the best next step from any point to any point in a network.
- **Gradient** — the derivatives that let an AI *learn*.
- **Trigonometry** — sine, cosine, and the geometry of space.

Each domain is a module — added to the machine like a new cartridge of knowledge. The library grows; the hardware stays the same. A new domain is a new ability, with no new chip.

3. One engine, two gifts

The OmniCube gives the machine two powers from a single design:

Direct answers. For any question with a *bounded* set of possibilities — arithmetic, a cipher, an activation function, a color conversion — the answer is simply *there*, retrieved in one step, for almost no energy. The computer often **already knows**.

An engine for everything else. The same Math OmniCube that returns $a \times b$ in one step is also the **multiply-and-add engine** behind artificial intelligence, physics, and graphics — the operations those fields are built from. So the OmniCube is not only a library; it is the arithmetic heart that drives the AI matrix engine and the science behind it.

4. The honest rule that keeps it real

There is a discipline at the center of the OmniCube, and it is what keeps every claim in this work grounded:

We pre-store only what is finite. We never pretend to store infinity.

Some things are bounded — there are only so many ways to multiply two bytes, only so many pixels, only so many cipher blocks. Those we store, completely. But some things are unbounded — the next word in an open-ended sentence, the factors of an enormous number. Those can never be fully written down, by anyone, ever; there is not enough matter in the universe. For those, the OmniCube does not pretend to *know* the answer — it **accelerates the math** that finds it, one operation at a time. Knowing the difference is what separates a real machine from a fantasy, and TimeLightHub is built on the real side of that line.

5. Why it matters

A computer that often *already knows the answer* spends almost nothing to give it. A computer whose knowledge is a **growing library of swappable modules** gains new abilities without new hardware. And a computer honest about the line between the finite and the infinite is one you can trust.

The OmniCube is the memory of everything TimeLightHub knows — written in light, kept without power, and reached in a single step. The geometric elements are the body; the OmniCube is the mind's library. Together, they are a computer that thinks in light.

Samer Beyrouti — TimeLightHub — June 2026 — Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151 The full domain catalogue and the factory-writing process are shared with partners under agreement.

The Architecture

Thirteen Instruments, and the Orchestra They Make

Samer Beyrouti — Inventor & Architect · TimeLightHub · June 2026

A piano is one instrument. An orchestra is a hundred — and it can play anything. TimeLightHub is built the second way: thirteen small photonic elements that, arranged differently, become whatever machine the task needs.

1. A kit, not a chip

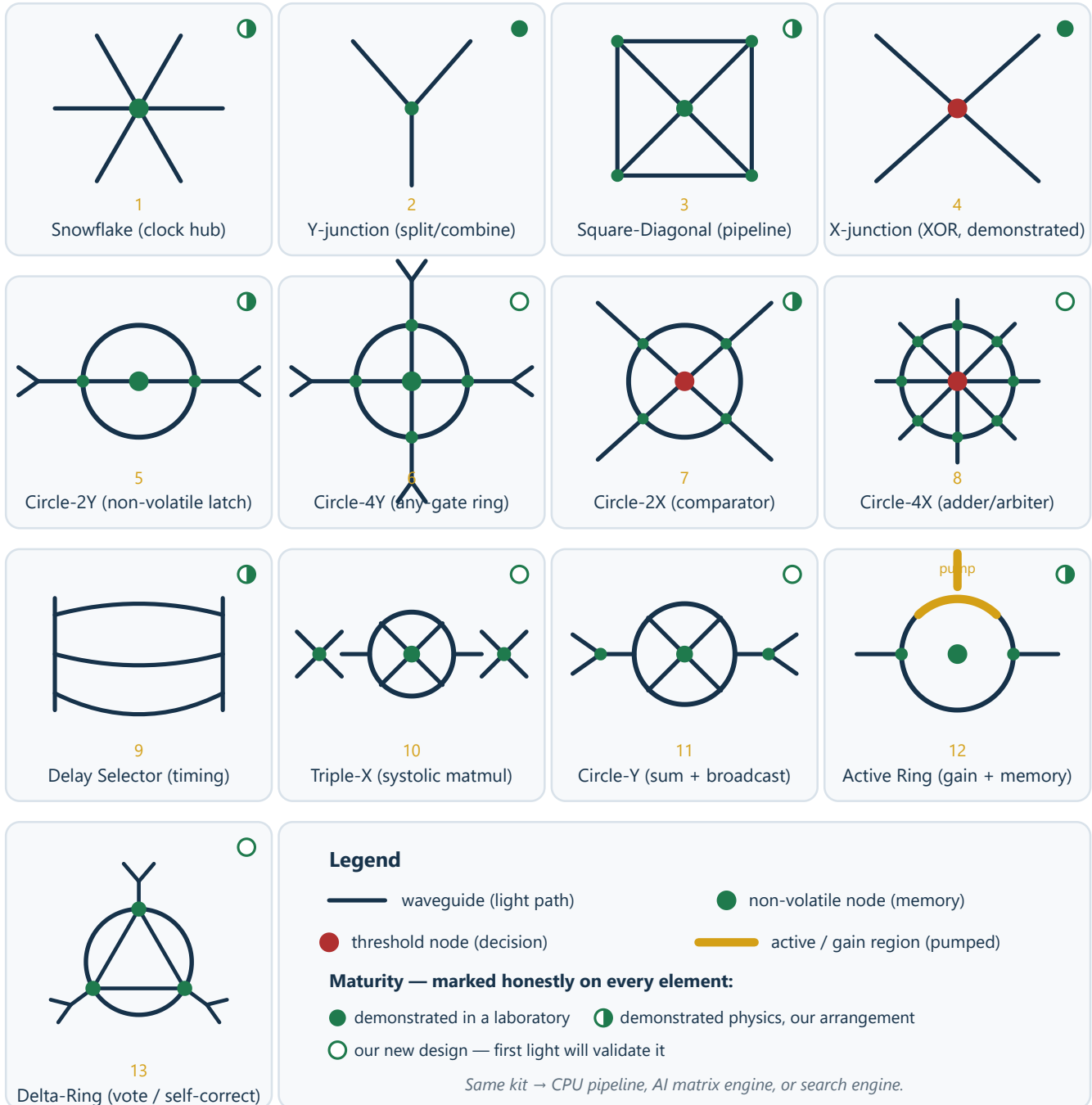
Most computers are frozen. Their wiring is fixed at the factory, and a processor is a processor forever. TimeLightHub is different by design. It is a **kit of thirteen photonic elements**, and the machine you get depends on how you arrange them — and the arrangement can be changed by the color of the light you send in.

This is the architecture's first principle: **arrangement is function**. The same elements, in a line, are a processor's pipeline. In a grid, an artificial-intelligence engine. In a tree, a search engine. The hardware does not change; the music does.

2. The thirteen elements

TimeLightHub — The 13 Photonic Elements

a composable kit: the same elements, arranged differently, become different machines



They fall into seven sections, like an orchestra:

The conductor. The **Snowflake** sends one pulse of light — the heartbeat — to every element at once, along arms of equal length. Because every path is matched by construction, the clock arrives everywhere with near-zero skew — not tuned into correctness, but *built* correct.

The memory. The **Circle-2Y latch** is a ring of light with a switchable node at its center that holds a bit *with the power off* — a non-volatile memory made of light. The **Active Ring** adds gain: it is memory and amplifier at once, keeping signals bright as they travel.

The logic. The **X-junction** crosses two beams to compute the exclusive-or — already demonstrated in the laboratory in a sixth of a trillionth of a second. The **Square-Diagonal** combines and times signals by the geometry of their paths. The **Circle gates** (4Y, 2X, 4X) provide the full set of logical operations and a reconfigurable gate that becomes any function by changing the light.

The routing. The **Y-junction** splits, combines, and broadcasts. The **Delay Selector** times signals to the picosecond by choosing the length of the path they travel.

The compute fabric. The **Triple-X cell** is the engine of artificial intelligence — a matrix multiplier with its weights held in place. The **Circle-Y cell** sums, activates, and broadcasts: the exact motion of a neural-network layer.

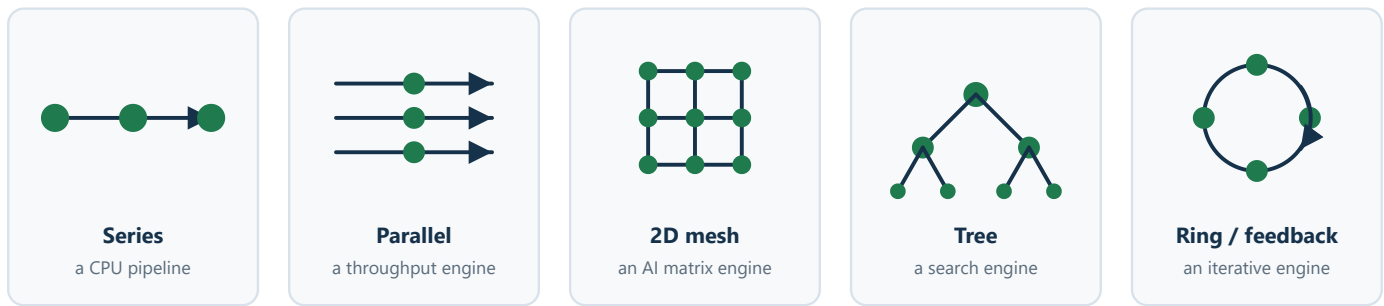
The self-correction. The **Delta-Ring** — a triangle of light inside a circle — takes three copies of a value and **votes**, discarding noise and error by agreement. The most reliable machines in the world, in spacecraft, work this way; here it is built into the optics.

And a maturity mark on every instrument. The catalog carries it openly: a filled circle for elements **demonstrated in a laboratory** (the X-junction's exclusive-or, the Y-junction's split), a half circle for elements built from **demonstrated physics in our new arrangement** (the latch, the Active Ring, the Snowflake), and an open circle for **our own new designs** — the Triple-X, the Circle-Y, the Delta-Ring, the any-gate ring — whose first validation comes at first light. An honest orchestra knows which players have already performed.

Each element also replaces a specific piece of electronic hardware — a logic gate, a register, a filter, a voter, a buffer — but with almost no heat, no power to remember, and no wires to charge. The full physics of each is shared with partners under agreement.

3. The orchestra: one kit, many machines

Same kit, different machine — arrangement is function



Wire the elements one way and they become one machine; wire them another and they become a different one entirely:

- **In series** — a deep pipeline, for complex step-by-step programs: a **CPU**.
- **In parallel**, each on its own color — a wide engine that does many things at once: a **throughput machine**.
- **In a grid** — data flowing through stored weights: an **AI matrix engine**.
- **In a tree** — branching to compare and reduce: a **search engine**.
- **In a loop** — feeding its own output back: an **iterative engine**, for simulation.

A CPU, a GPU, and a search engine are not three chips here. They are the same thirteen elements, arranged differently. And because each element responds to color, **the arrangement itself can be reprogrammed by light**.

4. LUMINA: the score

An orchestra needs a score, and a conductor to read it. In TimeLightHub that is **LUMINA**, the programming language we have specified for the machine. A LUMINA program is compiled not into electronic instructions but into a **schedule of light** — which colors fire, how brightly, how long, and in what order — that arranges the elements for the task at hand.

To make it concrete, here is what one layer of a neural network looks like in LUMINA — five lines, each mapping onto an element you have just met:

```
W = load_weights("encoder", 1)    // written once into the cells – they never move
y = matmul(x, W)                  // light flows through the Triple-X grid
y = @activation(y, GELU)          // one lookup in the AI-activation domain
```

```
y = vote3(y, y2, y3)           // the Delta-Ring discards any disagreeing copy
emit(y)                         // the Circle-Y broadcasts to the next layer
```

The compiler turns this into the score — a timed schedule of laser pulses. The language is designed and specified; its compiler and a physics-honest simulator are the first software deliverables, and they need no hardware at all to begin. The same hardware, given a different score, plays a different piece.

5. Why this matters

A composable, reconfigurable computer made of a handful of light-elements is something new. It is smaller, because light does more with fewer parts. It is cooler, because light does not heat its own wires. It remembers without power, and it corrects its own mistakes. And it is not one machine but many, waiting in the same hardware for the right score.

The instruments are designed. The next papers show them one by one — and then what they play: the AI engine where the weights never move, the energy that barely registers, and the short, honest road to the first note.

Samer Beyrouti — TimeLightHub — June 2026 — Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151

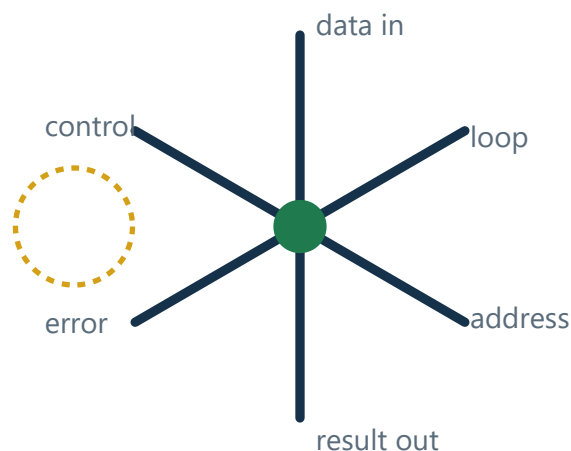
The Element Dossier

The Thirteen Instruments, One by One

Samer Beyrouti — Inventor & Architect · TimeLightHub · June 2026

Here is every element, drawn and explained on its own. Each is a small structure of light — and each replaces a piece of electronic hardware, but with almost no heat, no power to remember, and no wires to charge. And each carries an honest mark: ● already demonstrated in a laboratory, ○ demonstrated physics in our new arrangement, ○ our own new design, awaiting first light. The deeper physics of each is shared with partners under agreement.

1 · Snowflake — the conductor ○



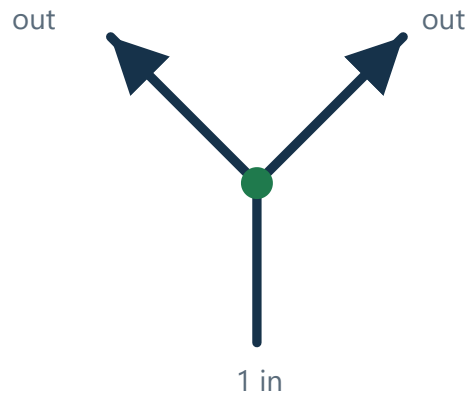
Six light arms around a hub (the dashed ring marks the heartbeat origin).
One pulse from the hub reaches every element at once, perfectly in time.

Replaces: the clock tree (with zero skew).

Six arms of light meet at a central hub. A single pulse — the heartbeat — leaves the hub and reaches every element of the machine at the same instant, because every path is the same length. It is the

clock of TimeLightHub, with near-zero skew by construction: the timing is not tuned into the machine, it is *drawn* into it. Path-matched optical distribution is established practice; making it the conductor of a computer is ours. **Replaces the electronic clock tree.**

2 · Y-junction — split & combine •

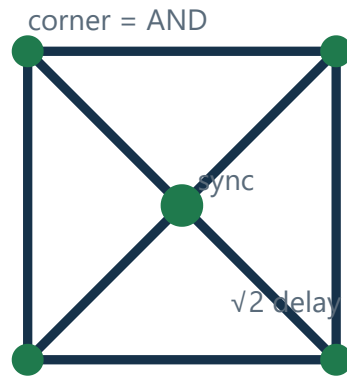


One beam splits to two (broadcast) — or two combine into one (a sum).

Replaces: fan-out buffer & multiplexer. The combiner is also how light adds.

The simplest element, and one of the most useful. One beam splits cleanly into two — to broadcast a signal to many places — or two beams combine into one, which is also how light *adds*. Demonstrated, standard, reliable. **Replaces the fan-out buffer and the multiplexer.**

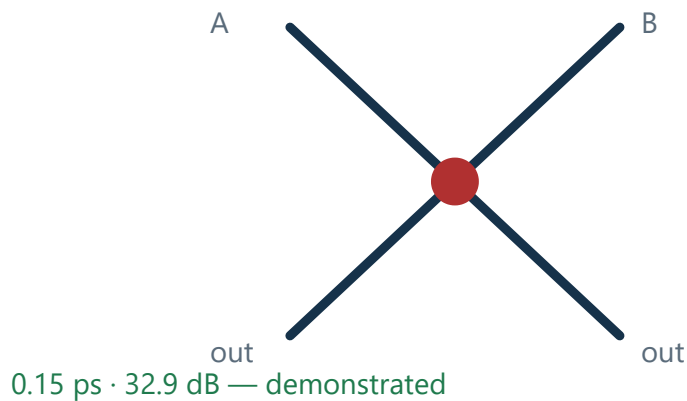
3 · Square-Diagonal — the pipeline ①



Corners detect coincidence (AND); the longer diagonal makes a built-in, zero-power timing delay. **Replaces: AND gate + pipeline register.**

A square of waveguide with crossing diagonals. Where signals coincide at a corner, it computes an AND; and because the diagonal is longer than the edge, it creates a precise, **zero-power timing delay** — the natural staging of a pipeline, built from geometry alone. Every primitive in it is demonstrated; the composed element is our design. **Replaces the AND gate and the pipeline register.**

4 · X-junction — XOR, demonstrated •

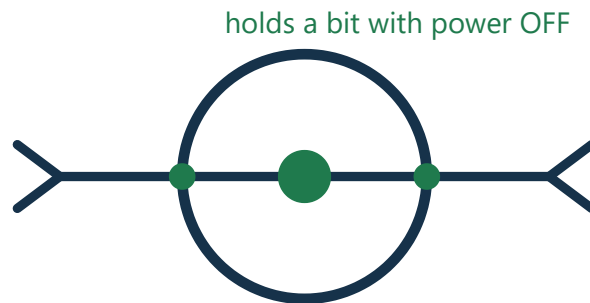


Two beams cross; interference gives XOR, AND, or a crossbar switch.

Replaces: the XOR gate + crossbar switch. The strongest, proven element.

Two beams cross, and their interference computes the exclusive-or — the workhorse of digital logic. This is the most solid element in the kit: it has already been demonstrated in the laboratory at **0.15 picoseconds with 33 dB contrast**. It also serves as an AND gate and a crossbar switch. **Replaces the XOR gate and the crossbar switch.**

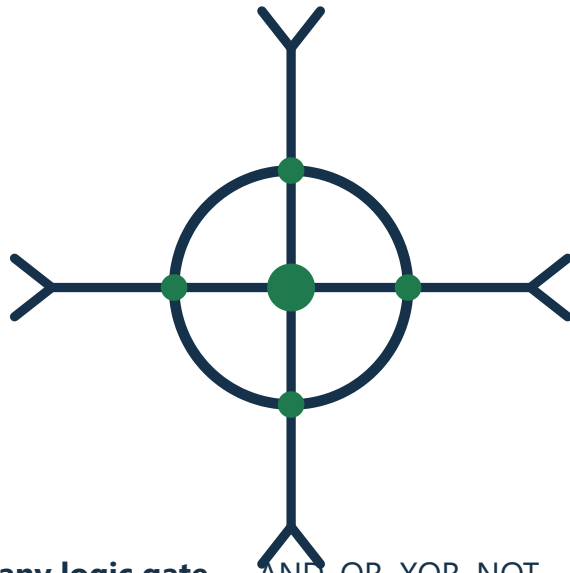
5 · Circle-2Y — the non-volatile latch ①



A ring with a light-switchable dot at its center stores a bit — and keeps it for years, unpowered. **Replaces: the flip-flop / register — now non-volatile.**

A ring of light with a switchable node at its center. The node holds a bit — and keeps it **with the power off**, for years, because the bit is stored as a stable molecular state. The storage physics is demonstrated in published research; our contribution is the latch itself — a non-volatile bit designed as a **composable element of a computer**, wired into the same fabric as the logic that uses it. **Replaces the flip-flop and register — now non-volatile.**

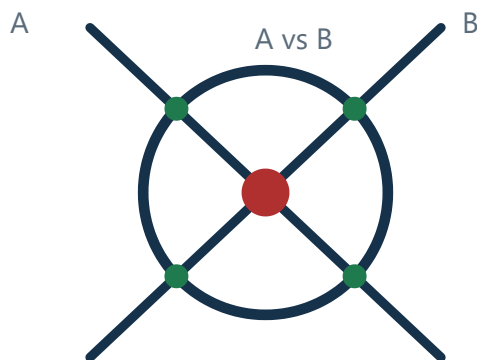
6 · Circle-4Y — the any-gate ring ○



One ring that becomes **any logic gate** — AND, OR, XOR, NOT — by changing the color and phase of its light. **Replaces: a reconfigurable logic block.**

The same ring, with four arms and counter-running waves inside. By changing only the color and phase of its light, it is designed to become **any logic gate you ask for** — AND, OR, XOR, NOT — with no change to the hardware. It is the boldest element in the kit and we say so: the integrated unit is unproven, inversion is the hard case, and the prototype will be its judge. **Designed to replace a reconfigurable logic cell (an FPGA element).**

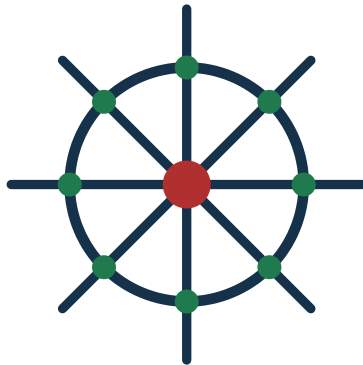
7 · Circle-2X — the comparator ①



Two paths through a ring compare two values and report which is larger. **Replaces: the comparator** (and a phase detector).

Two paths through a ring carry two values and bring them together, reporting which is larger. It is the decision-maker — the element that lets the machine choose. Built from demonstrated interference and threshold behavior, composed our way. **Replaces the comparator and phase detector.**

8 · Circle-4X — adder & arbiter ○

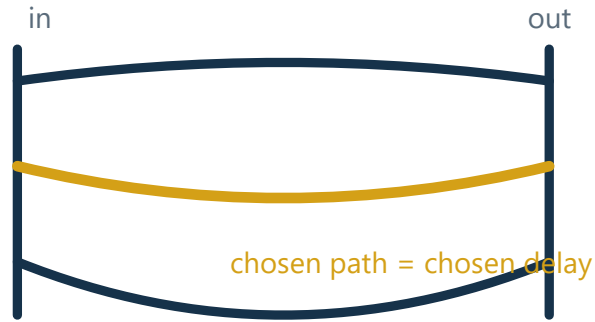


Four waveguides crossing = **eight ports** at a threshold node: adder + arbiter.

Replaces: ALU adder logic + a bus arbiter.

Eight ports meeting at a threshold node. Depending on how it is driven, it is a full adder, a priority arbiter that decides who goes first, or an eight-way logic gate — and it is the carry-keeper that lets byte-answers from the OmniCube compose into wide arithmetic. Our design, to be validated as a unit. **Designed to replace ALU adder logic and a bus arbiter.**

9 · Delay Selector — programmable timing ●



Pick which path the light takes, and you pick its delay — to the picosecond, by color. **Replaces: programmable delay lines & timing circuits.**

Several paths of different lengths between two points. Choose the path — by color — and you choose the delay, to the picosecond. Where the Square-Diagonal gives a fixed delay, this one is *adjustable*. Wavelength-selected paths are established art; the timing element is ours. **Replaces programmable delay lines and timing circuits.**

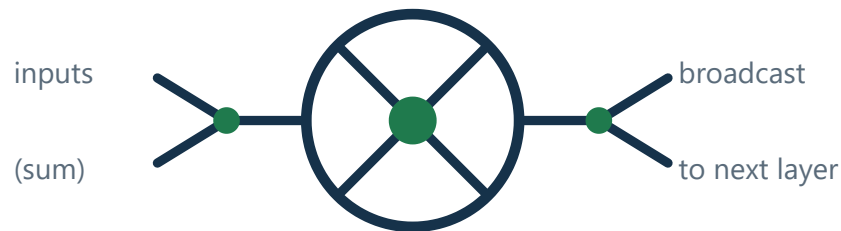
10 · Triple-X — the AI matrix cell ○



A weight sits in the center, stored and unmoving; data flows through and is multiplied as it passes. Tile these → an AI engine. **The weights never move.**

The engine of artificial intelligence. A weight is stored, unmoving, in the center; data flows in one side, is multiplied by the waiting weight, and flows out the other. Tile these into a grid and you have a matrix multiplier in which **the weights never move** — the secret to running AI without the energy of hauling the model around. This is our flagship design: every part of it rests on demonstrated physics, and the assembled cell is what the prototype exists to prove. **Designed to replace the heart of a GPU's matrix engine.**

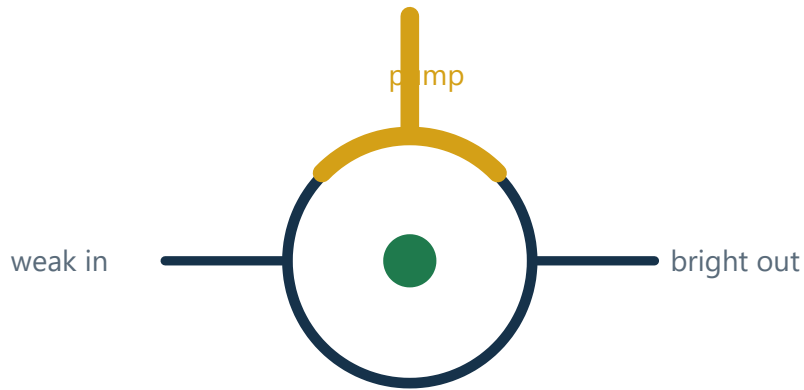
11 · Circle-Y — the neural cell ○



Adds up its inputs, applies the activation, and broadcasts the result onward — the exact motion of one neural-network layer. **Sum → activate → fan-out.**

It sums its inputs, applies the activation, and broadcasts the result to the next layer — the exact three-step motion of a neural-network layer, in one element. Where the Triple-X multiplies, the Circle-Y **sums and sends on.** Together they are a complete AI layer. Our design, awaiting first light. **Designed to replace a neuron's worth of accumulate-and-activate hardware.**

12 · Active Ring — gain & memory ●

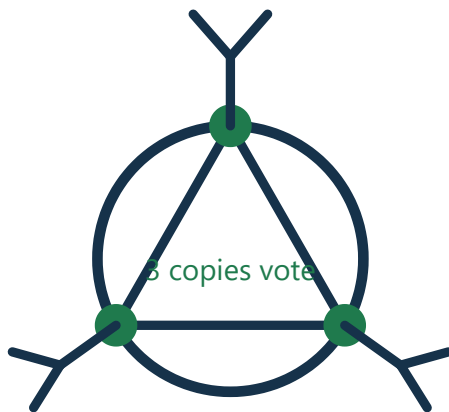


A pumped ring **amplifies, cleans, and re-times** the light — and stores a bit too. Memory and amplifier in one. **This is how loss is undone.**

A ring that is fed energy by a pump, so it can **amplify** the light passing through, **clean** away noise, and **re-time** the signal — and store a bit as well. Optical gain rings and all-optical regeneration are demonstrated, mature physics; the fusion of memory and amplifier in one element is ours. It is how TimeLightHub undoes the light it loses over long paths — and it is also where we spend our honesty: the pump is the machine's main energy and heat line-item, budgeted openly in the roadmap.

Replaces optical amplifiers and signal-regeneration hardware.

13 · Delta-Ring — vote & self-correct ○



Three copies of a value meet and **vote** — the majority wins, errors are thrown out.

Built-in fault tolerance — the way spacecraft computers stay reliable.

A triangle of light inside a circle, with a node at each of its three corners. It takes three copies of a value and lets them **vote** — the majority wins, and any error is thrown out. This is how the most reliable computers in the world, aboard spacecraft, stay correct under stress — and here it is designed directly into the optics. It is also our designed answer to the hardest open question in any analog machine: keeping values precise as they cascade. Our newest element, and not our last. **Designed to replace fault-tolerant voter circuits and a median filter — adding self-correction for free.**

The kit, together

Thirteen elements: a clock, two kinds of memory, five kinds of logic, routing and timing, two AI cells, an amplifier, and a self-corrector. Each is small, cool, and non-volatile. Two are already demonstrated; six stand on demonstrated physics in new arrangements; five are our own new designs — and the kit is open, with more elements in progress. None of them, alone, is a computer. But arranged together — in series, in a grid, in a tree — they become one: a CPU, an AI engine, a search engine, reconfigured by light.

That is TimeLightHub: not a chip, but an orchestra.

Samer Beyrouti — TimeLightHub — June 2026 — Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151 Full element physics and fabrication detail available to partners under agreement.

The AI Engine

Where the Weights Never Move

Samer Beyrouti — Inventor & Architect · TimeLightHub · June 2026

The secret of running artificial intelligence efficiently is not faster arithmetic. It is to stop carrying the model back and forth. TimeLightHub keeps the model still, and lets the light come to it.

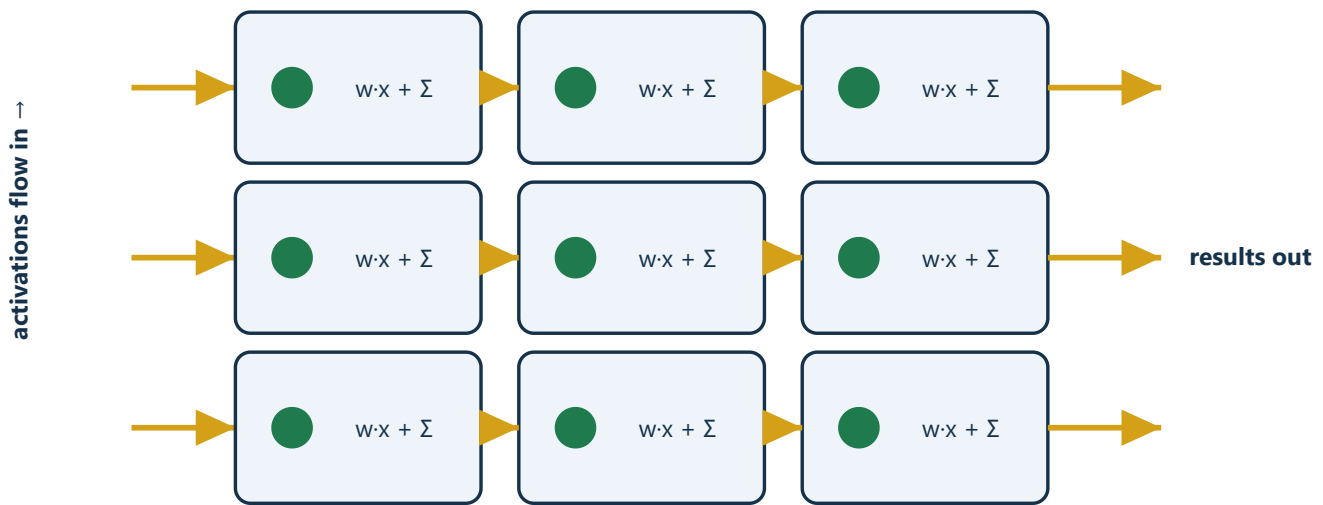
1. The real cost of AI

When an AI model writes a word, it does a staggering number of multiplications — but that is not where the energy goes. The energy goes into **moving the model**. For every single word, a modern chip drags its billions of weights out of memory and across the silicon to the place where the math happens, then does it again for the next word, and the next. The arithmetic is cheap. The *commute* is what costs.

This is why a high-end AI chip draws hundreds of watts: most of it is spent moving data, not computing.

2. The TimeLightHub answer: hold the weights still

The AI engine: the weights never move



● green = a stored weight — it stays in its cell; the light flows past it.

In a GPU the weights are hauled across the chip for every word — the biggest energy cost.
Here the weights live in the array; only the light moves.

TimeLightHub turns the problem inside out. Instead of carrying the weights to the math, it **stores each weight, permanently, in the very cell where the math happens** — and lets the data flow through.

The engine is a grid of **Triple-X cells**. Each cell holds one weight, stored in non-volatile light-memory that needs no power to remember. The activations — the live data — flow **in from one side, through the grid, and out the other**, and as they pass each cell, the light is multiplied by the weight waiting there and added to the running total. The answer emerges on the far side.

The weights never move. They sit in their cells, and only the light travels. The biggest energy cost in all of AI — moving the model — is simply gone.

3. The full layer, played by the orchestra

A neural-network layer is a weighted sum, then an activation, then a hand-off to the next layer. The TimeLightHub orchestra plays it directly:

- The **Triple-X grid** does the weighted sum — the matrix multiply — with weights in place.
- The **Circle-Y cells** add up the results and apply the activation, then **broadcast** the output to the next layer.

- The **Active Rings** keep the light bright and clean as it flows.
- The **Delta-Rings** vote among copies and discard noise, so errors do not build up layer after layer.

Each layer uses different weights — but because the weights live in the cells, **there is nothing to reload**. In an ordinary optical accelerator, swapping the model for each layer is the bottleneck. Here there is no swap: the model *is* the machine.

And we are honest about scale: a model is only "the machine" up to the number of cells the machine has. A small model lives whole inside one TLH Unit; a large one tiles across many Units, the way large models already shard across many GPUs today — with one difference in our favor: once placed, the shards *stay* placed, in memory that costs nothing to keep. The industry's move to 4-bit weights is timing on our side too — every halving of weight width doubles the model a given fabric can hold.

4. How big is the win

For artificial intelligence, the number that matters is **energy per answer** — the energy to produce one word — because that is the electricity bill and the cooling bill. The whole of it comes down to one quantity: **how little energy it takes to read one weight** from light-memory into the math.

The arithmetic of the claim is simple, public, and checkable. Generating one word is memory-bound: every weight is read once. Published photonics has already shown the multiply itself can cost almost nothing — less than one photon per operation. So the bill is the weight-read. **If reading one stored weight costs a few tenths of a picojoule, the energy per word lands tens of times below today's flagship AI chip — even after paying the lasers and amplifiers honestly. At a few picojoules the advantage narrows; above ten, it is gone.** That window is wide, it is physical, and it is measurable.

That single quantity is the first thing the prototype will measure, and it is the honest heart of the claim. One boundary we draw ourselves: this is a win in **energy per answer**, not raw throughput — a rack of top GPUs will still produce more words per second than a first TimeLightHub. It will pay many times the energy for each of them.

5. What this opens

An AI engine that holds its model still is not only more efficient — it is cooler, quieter, and non-volatile, ready to answer the instant it is switched on, with the model already in place. It is the natural home for energy-constrained intelligence: in data centers that can no longer afford the heat, and at the edge, where the power comes from a battery.

The model stays. The light moves. That is the engine.

Samer Beyrouti — TimeLightHub — June 2026 — Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151

The Energy Advantage

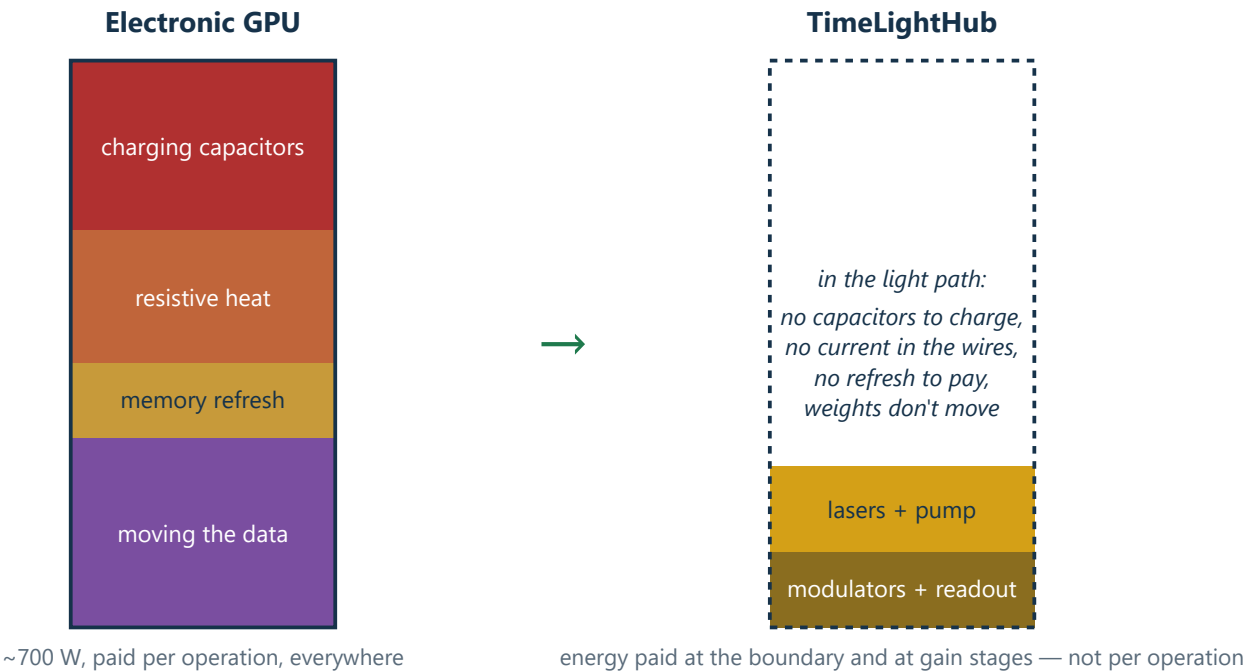
A Computer That Barely Spends

Samer Beyrouti — Inventor & Architect · TimeLightHub · June 2026

The cheapest computation is the one that never has to fight physics. TimeLightHub is built to avoid the four things electronics pays for every second.

1. The four taxes of electronics

Where the energy goes



Illustrative comparison, not a measurement. The whole advantage reduces to one number — the energy to read one stored weight — and that number is exactly what our first experiment measures.

An electronic computer spends its energy in four ways:

1. **Charging capacitors.** Every time a transistor flips, it charges and empties a tiny capacitor. Billions of transistors, billions of times a second — this is the single largest cost.
2. **Heating its own wires.** Current through resistance becomes heat. This is why a powerful chip needs hundreds of watts and constant cooling.
3. **Refreshing memory.** Ordinary memory forgets unless it is rewritten thousands of times a second — power spent just to *not forget*.
4. **Moving data.** The largest cost in artificial intelligence: hauling information across the chip, again and again.

Add them up, and most of a computer's energy is spent on everything *except* the answer.

2. How light avoids each one

TimeLightHub was designed, element by element, to not pay these taxes:

- **No capacitors to charge.** Computation happens in light passing through waveguides. A photon carries its information without charging anything. The largest electronic cost simply does not exist in the light path.
- **No current in the wires.** Light drives no current through resistance, so the compute path leaves almost no heat behind. What little the optics absorbs or scatters is a whisper next to a wire's roar — and the machine that barely heats barely needs cooling.
- **No memory to refresh.** Data is held as a stable molecular state that keeps itself, for years, with no power. Remembering is free.
- **No data to move.** In the AI engine, the weights live in the cells where the light reads them. The biggest cost of all is engineered away.

What remains is the computation itself — and the optical multiply is so efficient it has been performed, in the laboratory, with **less than one photon of light**. This is efficiency at the level of the laws of physics, not merely of better engineering.

3. Where the energy is actually spent — honestly

TimeLightHub is not magic; it does spend energy, in three places, all at the **edge** of the machine or at **spaced intervals** inside it:

- the **lasers** that create the light,
- the **modulators and readout** that write data onto the light at the input and read answers off it at the output, and

- the **pumps** that feed the sparse amplifying rings restoring light over long paths — the largest of the three, and the one we budget most carefully.

The key is *where* this energy sits. Electronics pays its four taxes **for every operation, everywhere**. TimeLightHub pays at the boundary, and at regeneration stations spaced like substations on a power grid — not per multiply, not per word. That is the structural reason the energy per answer can be so much lower. And it is also where our two open engineering questions live: how deep an analog value can cascade before precision fades, and what the pumps truly cost in energy, noise, and heat. Both have designed answers — the Delta-Ring's voting, the passive re-cleaning thresholds, pumping at intervals — and both are measured, not assumed, on the roadmap.

4. The honest measure

For artificial intelligence, the meaningful number is **energy per answer**. Because the model's weights never move and the compute path makes almost no heat, that energy is dominated by one quantity: how little it costs to read a weight from light-memory. The arithmetic is public and checkable: **at a few tenths of a picojoule per weight read, the energy per word lands tens of times below today's flagship AI chip; at a few picojoules the advantage narrows; above ten it disappears**. The window is wide and it is physical.

We do not ask you to take this on faith. We built an independent, physics-honest simulator of the architecture — one that charges every real cost (the light that is lost, the noise the amplifiers add, the energy each pump and each readout spends) and refuses to assume anything is free. Built bottom-up from first principles, it reconstructs the same energy budget and confirms the advantage holds across the whole plausible range of that one number. The result is a conditional, stated plainly: deliver the weight read cheaply enough, and the energy per answer lands far below today's flagship chip. And the claim reduces to a single number that the very first hardware experiment measures directly.

5. What a low-energy computer means

A computer that barely spends changes what is possible. Data centers, now limited by heat and power as much as by speed, could do far more in the same budget. Intelligence that today needs a wall socket could run on a battery. And a machine that wastes almost nothing is, by its nature, smaller, quieter, and longer-lived.

That is the promise of computing in light: not just faster answers, but answers that cost almost nothing to give.

Samer Beyrouti — TimeLightHub — June 2026 — Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151

The Honeycomb

How the Elements Become a Machine

Samer Beyrouti — Inventor & Architect · TimeLightHub · June 2026

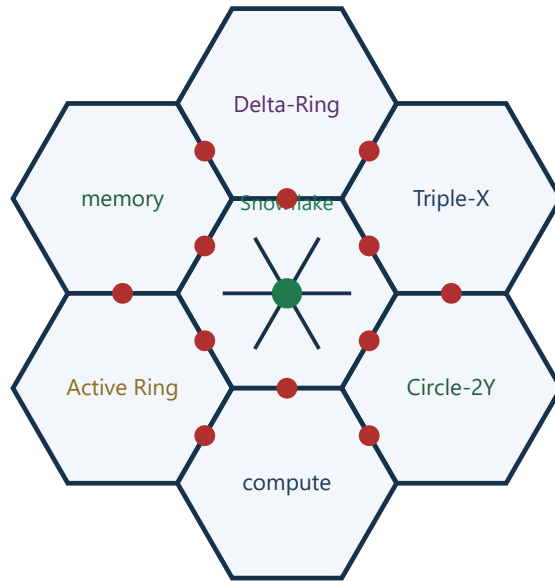
A bee builds in hexagons because it is the most efficient shape in nature — the most space, the least wall, the least wax. TimeLightHub is built the same way. The honeycomb is where the thirteen elements stop being a kit and become a machine.

1. Why a honeycomb

The elements have to live somewhere — tiled across a surface, wired to their neighbours. The shape of that surface is not a detail; it decides how much light is lost, how far signals travel, and how densely the machine can think. We chose the hexagon, for three reasons that are all real and provable.

The Honeycomb Fabric — how the elements tile

hexagons pack with the shortest walls (least loss); each cell touches 6 neighbours; a dot on every shared edge



- edge-dot: passive threshold (re-cleans the value at each hop)
scales in rings: 6 to 12 to 18 to ...

- **It wastes the least.** It is a proven fact of mathematics (the honeycomb theorem) that hexagons tile a surface with the **shortest possible walls** for a given area. For us, "wall" means waveguide — so a honeycomb gives the **shortest connections between cells, the least lost light, and the densest packing**. Nature reached this answer for the same reason.
- **It matches the material.** The photonic crystals that TimeLightHub's light flows through are **themselves hexagonal** — it is the natural lattice of the medium. The fabric and the material agree.
- **It matches the conductor.** The Snowflake has **six arms**. Place one at the centre of each hexagon and its arms point exactly at its **six neighbours**. The clock and the fabric become a single structure, with no waste.

Each cell touches six others — more than a square grid's four — which means shorter paths, more ways to route, and a natural ability to **flow around a failed cell**. And on every shared wall between two cells sits a **dot**: a small node where the two cells meet.

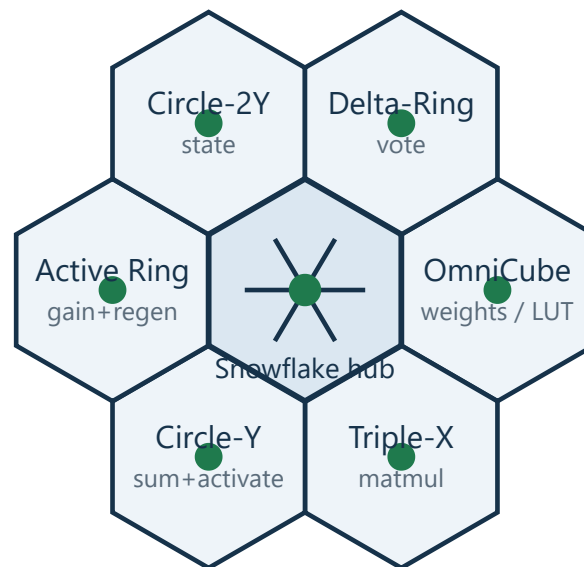
2. The dots that keep the light clean

Those edge-dots do quiet, important work. Most of them are **passive** — each one gently **re-cleans the value** as it hops from one cell to the next, with no power at all, so small errors never pile up across the fabric. A few of them, spaced out like substations on a power grid, are **active** — they give the light a boost of energy where it has travelled far and grown faint. Mostly passive, sparingly active: the fabric keeps its own signals strong and clean, and it does so cheaply. This is also our designed answer to the two questions every analog machine must face — precision over distance, and the cost of gain — and the roadmap measures both.

3. The TLH Unit — the complete machine

The honeycomb is how the pieces become a whole. A **TLH Unit** is the complete thing: the honeycomb fabric, populated with the elements and the OmniCube, with its memory and its light-based input and output — a working computer.

The TLH Unit — one hub + six sockets (an AI core)



Swap the sockets and the same unit becomes a CPU, a memory bank, or a search engine.

Tile units in rings: 6 to 12 to 18 cells.

The smallest piece of it is a **core**: one Snowflake hub surrounded by six cells, each a **socket** that holds any element or an OmniCube. And here is the beauty of it — *the same core becomes a different machine depending on what you put in the sockets*. Fill them one way and it is a CPU core; another way, an AI core; another, a block of memory; another, a search engine. The hardware does not change — the music does. Tile these cores across the honeycomb, ring by ring — six cells, then twelve, then eighteen — and the cores grow into a full Unit, the way a multi-core chip is built, but every core is a flower of light.

4. Built in layers — the Unit in three dimensions

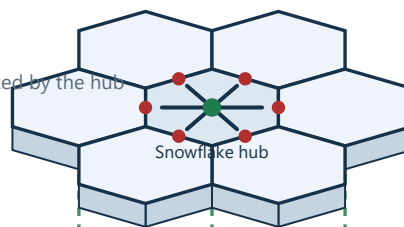
A honeycomb need not be flat. Stack the layers, and each cell gains neighbours **above and below** as well as around — eight ways for light to flow instead of six.

The TLH Unit in 3D — stacked layers of honeycomb light

each cell touches six neighbours in its plane, plus one above and one below — eight ways for light to flow

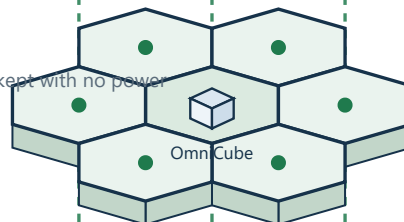
Compute layer

the element kit — logic, routing, AI cells, clocked by the hub



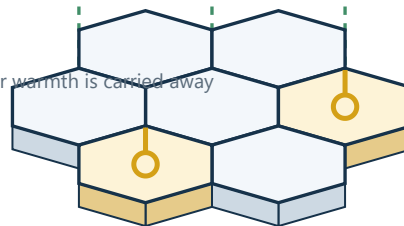
Memory layer

non-volatile latches + the OmniCube — data kept with no power



Gain layer

sparse pumped boosters — placed where their warmth is carried away



vertical couplers —
light passes between layers

A design, drawn true to the physics: layers separate jobs, the pumped layer is thermally isolated, and every layer-hop has a coupling cost we will measure — the densest way to pack thinking light into a space small enough to hold.

The stack is drawn the way it would be built, and each layer takes a job. The **compute layer** holds the element kit, clocked by its Snowflake hubs. The **memory layer** beneath it holds the latches and the OmniCube — the data sitting directly under the logic that uses it, one vertical hop away. And the **gain layer** holds the sparse pumped boosters, set apart **so their warmth can be carried away without touching the computation**. Vertical couplers pass the light between layers; each layer-hop has a coupling cost, and measuring it is on the roadmap — we draw the machine true to its physics, not to a wish.

A three-dimensional honeycomb is the densest way to pack thinking light into a small space — which is how a powerful TimeLightHub can be small enough to hold.

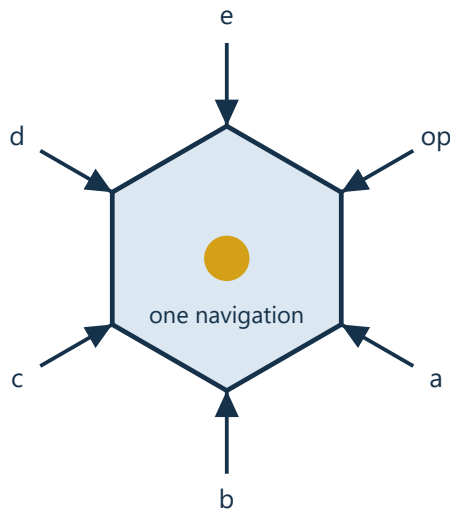
5. What the honeycomb makes possible

When compute, memory, and wiring all live in the same cells, something deep happens: the old boundaries between *processor*, *memory*, and *interconnect* dissolve. A honeycomb of light can be made into **any of them, anywhere, reprogrammed by colour** — and because the memory is non-volatile, the data can be worked on **right where it is stored**, so almost nothing has to move. A surface that computes where it remembers, corrects its own errors by agreement, and can be a CPU here and a search engine there — that is not a chip with parts. It is a single, living sheet of light. We are still discovering everything it can do.

6. A look ahead — the Hexacube

The OmniCube answers a question with two inputs — *a*, *b*, *operation* — in a single step. The hexagon invites a larger idea: a **Hexacube**, a lookup that takes **up to six inputs at once** (one per edge) and returns a single **fused** answer — several operations folded into one navigation.

The Hexacube — six inputs, one fused answer



More inputs, fewer bits each

2 ops x 8-bit	65K nodes	OmniCube
6 ops x 8-bit	3×10^{14} nodes	impossible
6 ops x 4-bit	16.7M nodes	feasible
4 ops x 4-bit	65K nodes	trivial

4-bit is the 2026 AI standard (FP4/INT4) — so a 6-input Hexacube is feasible, and does a fused multi-operand op in ONE step.

Fewer chained steps - less error build-up (helps precision) - hexagonal prism stacks in the 3D honeycomb.

There is an honest limit: more inputs mean fewer bits each, or the table grows beyond what any universe could hold. But here the timing is on our side — **artificial intelligence has just standardised on 4-bit numbers** (the 2026 frontier), and at four bits a **six-input Hexacube is entirely feasible**, doing in *one* step what now takes several. Fewer steps means less error and more speed, and its six-sided shape stacks perfectly into the honeycomb. We are keeping the Hexacube as a **clearly marked next horizon** — to be confirmed on a living prototype, where the surprises of an active fabric are the only true test.

7. The whole, at last

This is the picture complete: thirteen elements, an OmniCube of pre-written answers, and a honeycomb that tiles them into a cool, non-volatile, self-correcting sheet of computing light — small enough to hold, scalable ring by ring and layer by layer, and reshaped by colour into whatever machine the moment needs. The kit has become a Unit. The orchestra has its stage.

Hexacube design are shared in full with partners under agreement.

The Roadmap

From One Cell to the First Light

Samer Beyrouti — Inventor & Architect · TimeLightHub · June 2026

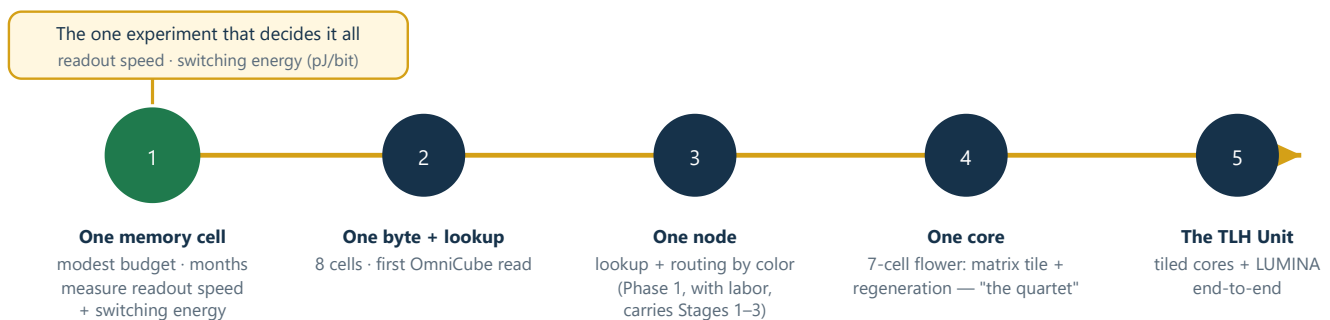
We would rather show a quartet that plays beautifully than promise a symphony on paper. The road to TimeLightHub is short at the start, and every step earns the next.

1. Measure first

A complete architecture is a beginning, not a proof. The honest way to build something new is to find the one experiment that decides everything, and do it first — cheaply, quickly, and in the open.

For TimeLightHub, that experiment is small and clear.

The road to first light — measure first, scale later



Each stage passes a go/no-go gate — did it do what the physics predicted? — before the next begins.

Honest total: a working multi-element prototype is realistically a multi-year program beyond Phase 1 — we say so plainly, because every stage stands on a measured result, not a promise.

2. The one experiment that decides it all

Build a single light-switchable memory cell, and measure two things: how *fast* it can be read, and how *little energy* it takes to write and read one bit.

These two numbers decide the whole architecture. The first tells us how quickly the machine can run; the second is the heart of the energy advantage. The cell is built from a well-studied light-switchable material, on a path drawn from published science. The cost is modest — **a well-bounded first experiment: months, not years; materials-scale, not program-scale; scoped with the partner** — and the result is **publishable on its own, whatever it shows**.

This is the first note. Everything else follows from it.

3. The staged path

Each stage earns the next, and none is begun before the last has proven itself:

1. **One cell** — write, read, reset; measure speed and energy. (*The decisive first step.*)
2. **One byte** — eight cells and the first OmniCube lookup: the machine's first *answer*.
3. **One node** — a full processing element: a lookup, a routing decision by color, a stored result. (*Our Phase-1 ask carries the program, with labor, through here — scoped with the partner.*)
4. **One core** — the seven-cell flower: a matrix tile multiplying with in-place weights, regeneration keeping it clean — the quartet, playing in tune.
5. **The TLH Unit** — tiled cores, the complete element set, and a LUMINA program running end to end.

At each gate we ask a simple question — *did it do what the physics predicted?* — and we proceed only if the answer is yes. And we say the full cost plainly: beyond Phase 1, a working multi-element prototype is realistically a multi-year program of a million dollars or more. We state that openly because every stage of it is gated by a measurement, not a promise — this is how a serious instrument is built: one section at a time, each one heard before the next is added.

One part of that instrument already exists. Before any glass is cut, we built the architecture's **physics-honest simulator** — software that models every element with its real loss, noise, and energy, runs LUMINA programs end to end, and reports an honest cost for each. It does not replace the hardware measurements; it tells us, in advance, exactly which numbers the hardware must hit and how the machine behaves once it does. Every stage above now begins with a prediction the silicon is asked to confirm — not a hope, a target.

4. The four numbers we are chasing

The entire architecture, for all its breadth, comes down to four measurements:

1. **Readout speed** — how fast a stored value can be read.
2. **Switching energy** — how little it costs to write and read one bit.
3. **Per-cell loss** — how many cells can be joined before the light must be refreshed.
4. **Amplifier quality** — how deep a computation can run before noise and pump energy set the limit.

Each of these is a number a laboratory can produce. The first two decide the energy claim; the last two decide how large the orchestra can grow. Together they turn an architecture into evidence. We know exactly what we are looking for — and that clarity is what makes the first prototype worth building.

5. What we know, and what we don't — plainly

The architecture stands on published science, and most of its engineering challenges are answered by the element designs themselves: the weights that never move, the regeneration that undoes loss, the parallel width that feeds the machine. **Two questions remain genuinely open**, and we carry them forward in daylight:

- **Precision at depth.** Optical values are analog; how many operations can a value cascade through before small errors matter? Our designed answer is the Delta-Ring's voting, the passive re-cleaning thresholds on every cell wall, and robust few-level encoding — and the core stage (Stage 4) measures the true depth.
- **The pump budget.** The amplifying rings that keep light alive need pump energy, add noise, and make the machine's only real heat. Our designed answer is to pump at sparse intervals, not per operation, and to give the pumped layer its own floor of the 3D stack — and Stage 4 measures the watts and the noise directly.

A reader should know: a program that names its open questions is not a weaker program. It is one that knows exactly which experiments matter.

6. What we are looking for

This is an invitation to build. We are seeking:

- **A photonics laboratory or foundry** to make the first cell and the first node.
- **Research and funding partners** who see what a cool, non-volatile, composable optical computer could mean — for energy-efficient artificial intelligence, and for what comes after.

The architecture is complete. The physics of every element rests on published science. The intellectual property is protected, and the deeper engineering is ready to share with partners under agreement. What remains is the most exciting part: to send the first light through the first cell, and watch an idea become a machine.

7. The first light

There is a moment, in every new kind of computer, when the first signal runs through the first device and returns the right answer. For TimeLightHub, that moment is one memory cell, one beam of light, one measurement. It is close, it is affordable, and it is the only thing standing between a complete design and a working computer that thinks in light.

We are ready to take that step. We are looking for the partners to take it with us.

Samer Beyrouiti — TimeLightHub — June 2026 — Patent Pending — GB2615417.9

Connect@TimeLightHub.com · +961 3 097 151 Full technical documentation and the element-physics dossier available to partners under agreement.

TimeLightHub — Executive Summary

A composable photonic computer that thinks in light

Samer Beyrouti — Inventor & Architect · Connect@TimeLightHub.com · +961 3 097 151 · Patent Pending — GB2615417.9 · June 2026

The problem. Computing is hitting a wall made of heat and energy. A single high-end AI chip draws ~700 watts, most of it spent not on computing but on *moving data* and fighting heat. Memory leaks power just to remember. The industry needs a fundamentally more efficient way to compute — not a smaller transistor.

The idea. TimeLightHub computes with **light** instead of electricity, and remembers with light that **needs no power**. It is built from thirteen photonic elements that compose — like the sections of an orchestra — into a CPU, an AI engine, or a search engine, all from the same hardware, reconfigured by color.

Why it wins — energy. Electronic chips pay four taxes every second: charging capacitors, heating wires, refreshing memory, and moving data. TimeLightHub's compute path pays none of them. Light computes without charging capacitors or heating wires; its memory holds data with no power; and in its AI engine the model's **weights never move** — they live where the light reads them, eliminating the single largest energy cost in artificial intelligence. The whole claim reduces to one measurable number — the energy to read one stored weight — and published photonic physics puts the achievable window at **tens of times less energy per answer**. Our first experiment measures exactly that number.

What's new.

- A **non-volatile memory made of light** — keeps data with the power off, designed as a composable element of a computer (the storage physics is demonstrated science; the latch element is ours).
- An **AI engine where the weights never move** — answers each question for a fraction of the energy.
- **Self-correcting computation** — a three-fold element votes out errors, the way spacecraft computers do.
- A **composable element kit** — one device, many machines, reconfigured by light, with its own language (LUMINA) — every element honestly marked: demonstrated, grounded in

demonstrated physics, or our new design.

Honest status. This is a complete architecture whose every element rests on published science. It is not yet built, and two engineering questions (analog precision at depth; the amplifier pump budget) are stated openly, each with a designed answer and a stage that measures it. The entire design reduces to **one decisive first experiment**: build a single light-switchable memory cell and measure how fast and how cheaply it reads and writes — **a modest, well-bounded first experiment — months, not years; materials-scale, not program-scale; publishable on its own; budget scoped with the partner.** That result turns the design into evidence.

The market. Photonic computing is real and well funded (Lightmatter ~\$850M raised; Akhetonics, Celestial AI, Ayar Labs active). Every funded player builds *interconnects* or a *CPU*; **none pairs a non-volatile optical memory with a composable compute engine and a language.** That is TimeLightHub's space.

The ask. A photonics laboratory or foundry partner to build the first cell and node, and **support for Phase 1** — which carries the program, with labor, from the first cell through the first working node, scoped together with the partner. The intellectual property is protected; the deeper engineering is ready to share under agreement.

The vision. A computer of a few dozen parts instead of billions — that never loses your data, barely warms, corrects its own errors, and answers each question for a fraction of the energy. The science is real, the design is complete; the only question left is *who builds the first one*.

Contact: Samer Beyrouiti — Connect@TimeLightHub.com — +961 3 097 151 · Full technical papers and the element-physics dossier available to partners under agreement.

TimeLightHub — Pitch Deck

Speaker notes in italics. Each "---" is one slide. Figures live in ../diagrams/.

Slide 1 — Title

TimeLightHub

A computer that thinks in light

Samer Beyrouti — Inventor & Architect · Patent Pending — GB2615417.9 · June 2026

Connect@TimeLightHub.com · +961 3 097 151 *Open with the one line: "We designed a computer that runs cool, never forgets, and answers each question for a fraction of the energy."*

Slide 2 — The problem

Computing is hitting a wall made of heat and energy.

- A high-end AI chip draws ~700 watts — most of it spent *moving data* and fighting heat, not computing.
- Memory burns power just to remember.
- Smaller transistors won't fix a heat-and-energy problem.

The wall is real and the whole industry feels it.

Slide 3 — The idea

Compute with light. Remember with light that needs no power.

- 13 photonic elements that compose into different machines — a CPU, an AI engine, a search engine — from the same hardware.
- Reconfigured by color. One device, many computers.

- Every element honestly marked: demonstrated · demonstrated physics, our arrangement · our new design.

Show: `diagrams/element_catalog.svg`

Slide 4 — Why it wins: energy

Electronics pays four taxes. Our compute path pays none of them.

Electronic cost	TimeLightHub
Charging capacitors	light doesn't charge them
Heating wires	light drives no current
Refreshing memory	non-volatile — free to remember
Moving data	weights never move

What we *do* pay — lasers, modulators, amplifier pumps — sits at the boundary and at intervals, not per operation.

Show: `diagrams/energy_flow.svg`

If asked "how much better": the whole claim reduces to the energy of one weight-read; the physical window is tens-of-times better, and Stage 1 measures it.

Slide 5 — The breakthrough: AI where the weights never move

- LLM inference's biggest cost is hauling the model for every word.
- TimeLightHub stores each weight **in the cell where the light reads it**. The model is the machine.
- Result: each answer for a fraction of the energy. (The win is energy per answer — raw throughput still belongs to GPU racks, at many times the bill.)

Show: `diagrams/ai_engine_systolic.svg`

Slide 6 — What's new (and ours)

- **Non-volatile optical memory as a computer element** — keeps data with the power off. *The storage physics is demonstrated science; no funded competitor has built it into a composable compute fabric.*
- **Self-correcting computation** — votes out errors, like spacecraft computers.
- **A composable kit + a 3D honeycomb fabric + a language (LUMINA).**

Show: `diagrams/orchestra_topologies.svg` · `diagrams/tlh_unit_3d.svg`

Slide 7 — Honest status & the one experiment

A complete design, resting on published science — not yet built.

- The whole architecture reduces to **one experiment**: build a single memory cell; measure its read speed and switching energy.
- **A modest, well-bounded first experiment — months, not years; materials-scale, not program-scale; publishable on its own. Budget scoped with the partner.**
- Two engineering questions stay open (precision at depth; pump budget) — each named, each with a designed answer, each measured at a marked stage.

Show: `diagrams/roadmap.svg`

We'd rather show a quartet that plays in tune than promise a symphony on paper.

Slide 8 — Market

- Photonic computing is real and funded: **Lightmatter ~\$850M**, Akhetonics, Celestial AI, Ayar Labs.
 - They all build *interconnects* or a *CPU*. **None pairs non-volatile optical memory with a composable compute engine and a language.**
 - That gap is ours.
-

Slide 9 — The ask

A partner and a first step.

- A photonics lab or foundry to build the first cell and node.
 - **Phase 1** — carries the program, with labor, from the first cell through the first working node; scoped stage by stage with the partner.
 - Beyond Phase 1, we say it plainly: a multi-element prototype is a multi-year program — gated stage by stage by measured results.
 - IP protected; deeper engineering shared under agreement.
-

Slide 10 — The vision

A computer of a few dozen parts instead of billions. That never loses your data. Barely warms. Corrects its own errors. Answers each question for a fraction of the energy.

The science is real. The design is complete. The only question is who builds the first one.

Samer Beyrouti — Connect@TimeLightHub.com — +961 3 097 151